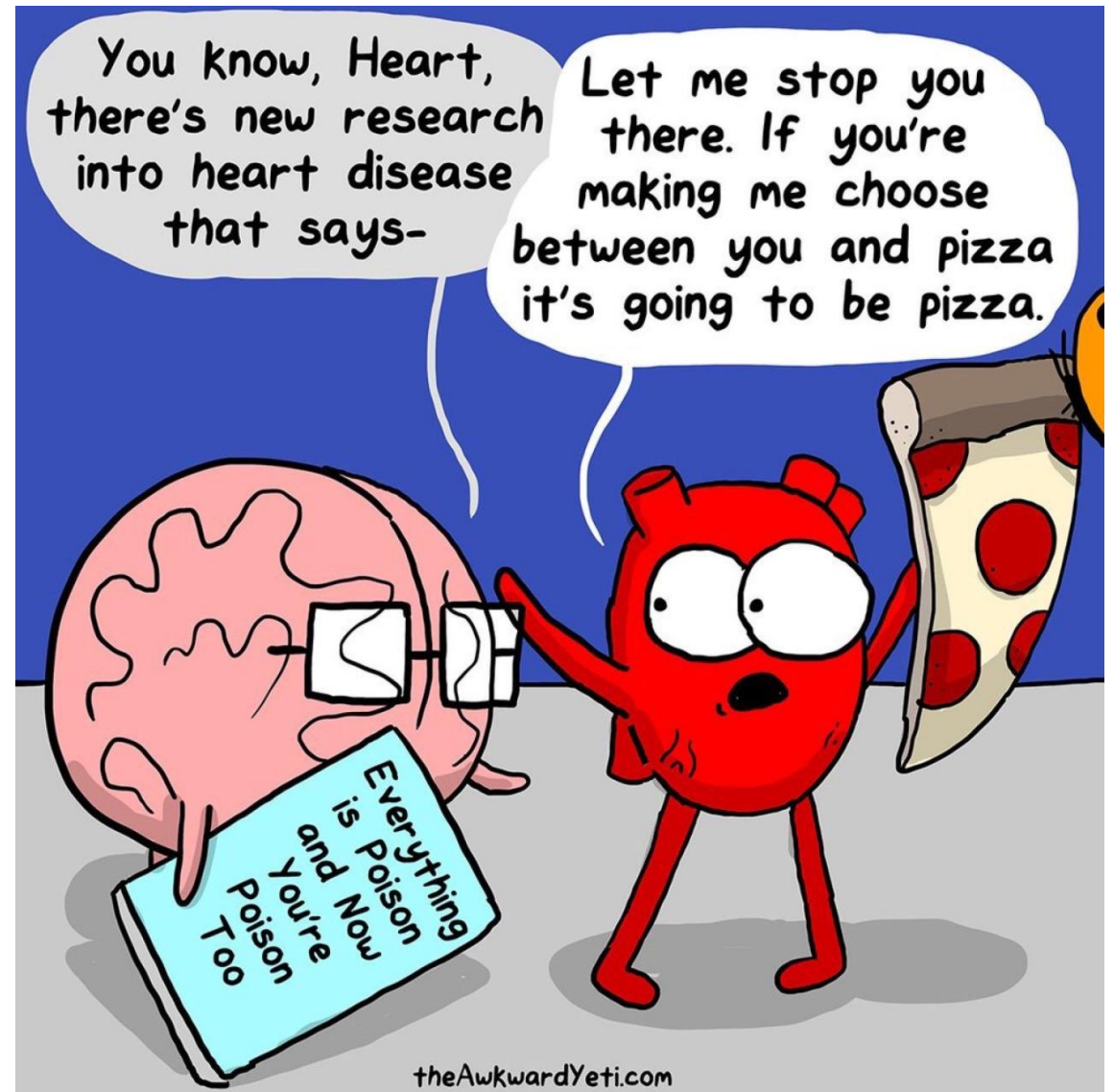


Knee Joint Mini-Review

T. Smith, PhD
(he/they)
COAR LE



Knee Joint

Tibiofemoral joint

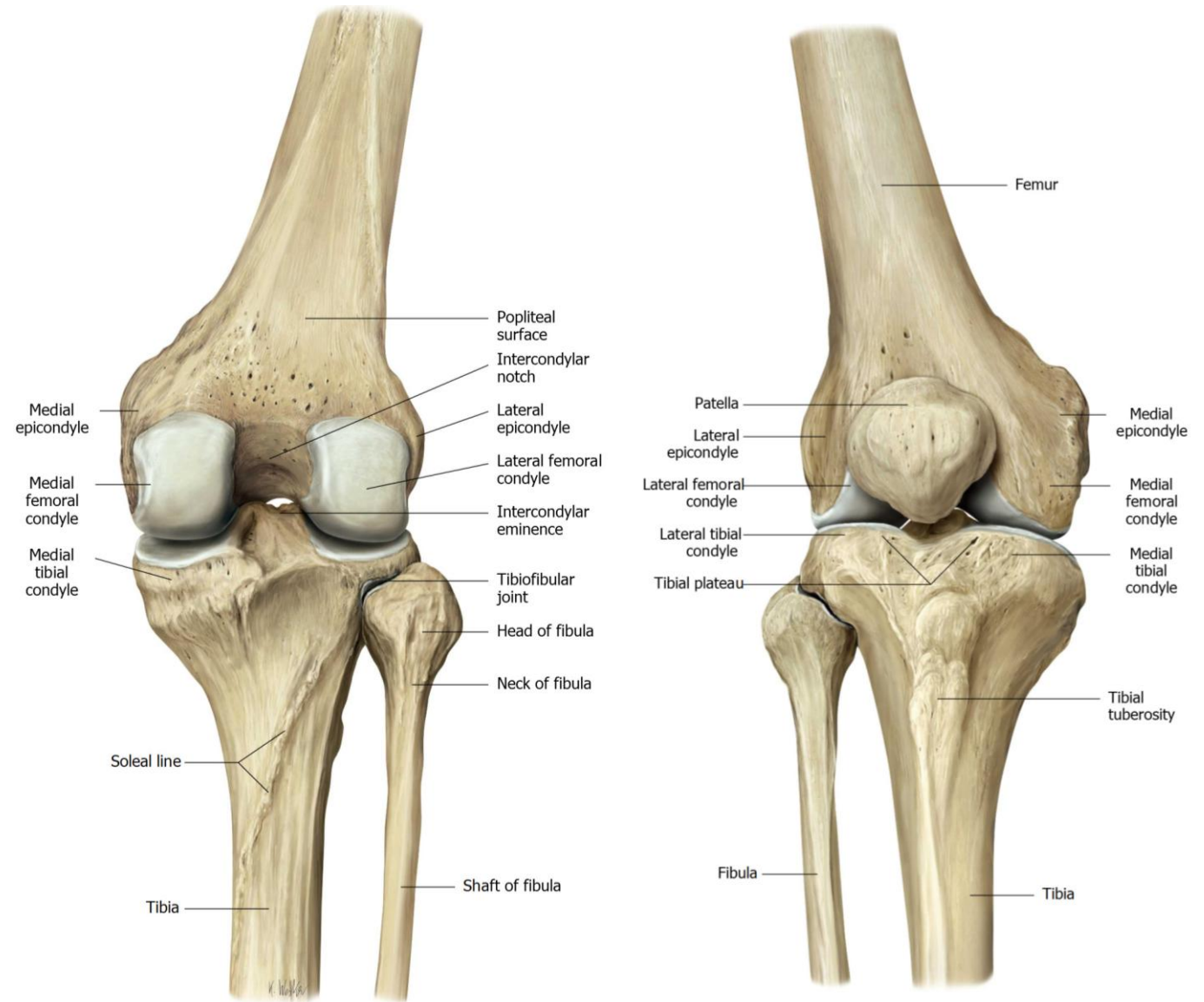
- bicondylar, synovial, hinge joint
- 1 degree of freedom: flex/extend
- load bearing surface between femoral condyles and tibial condyles

Patellofemoral joint

- patella = pulley for quadriceps
- increases mechanical advantage for knee flexion and extension

Tibiofibular joint

- nearby but not involved in knee joint action and is not load-bearing



Tibiofemoral Joint Support

Tibial condyles are not too deep
Even with menisci, joint congruence is not high

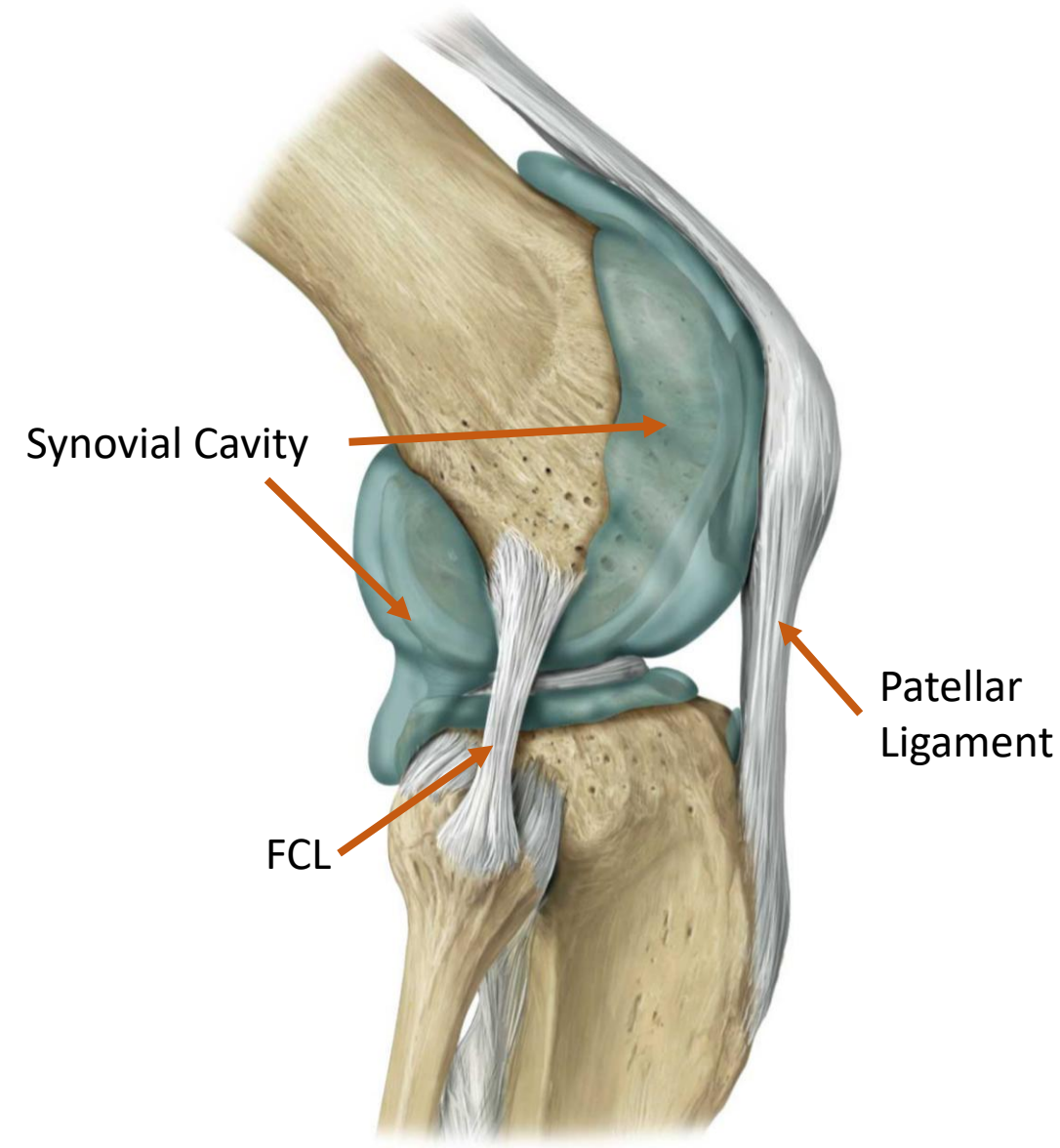
Ligament and Muscle support is key!

Extracapsular

- Patellar ligament
- Fibular collateral ligament
- Tibial collateral ligament
- Oblique popliteal ligament
- Arcuate popliteal ligament

Intracapsular

- Anterior cruciate ligament
- Posterior cruciate ligament
- Lateral/medial meniscus
- Coronary ligament of the knee



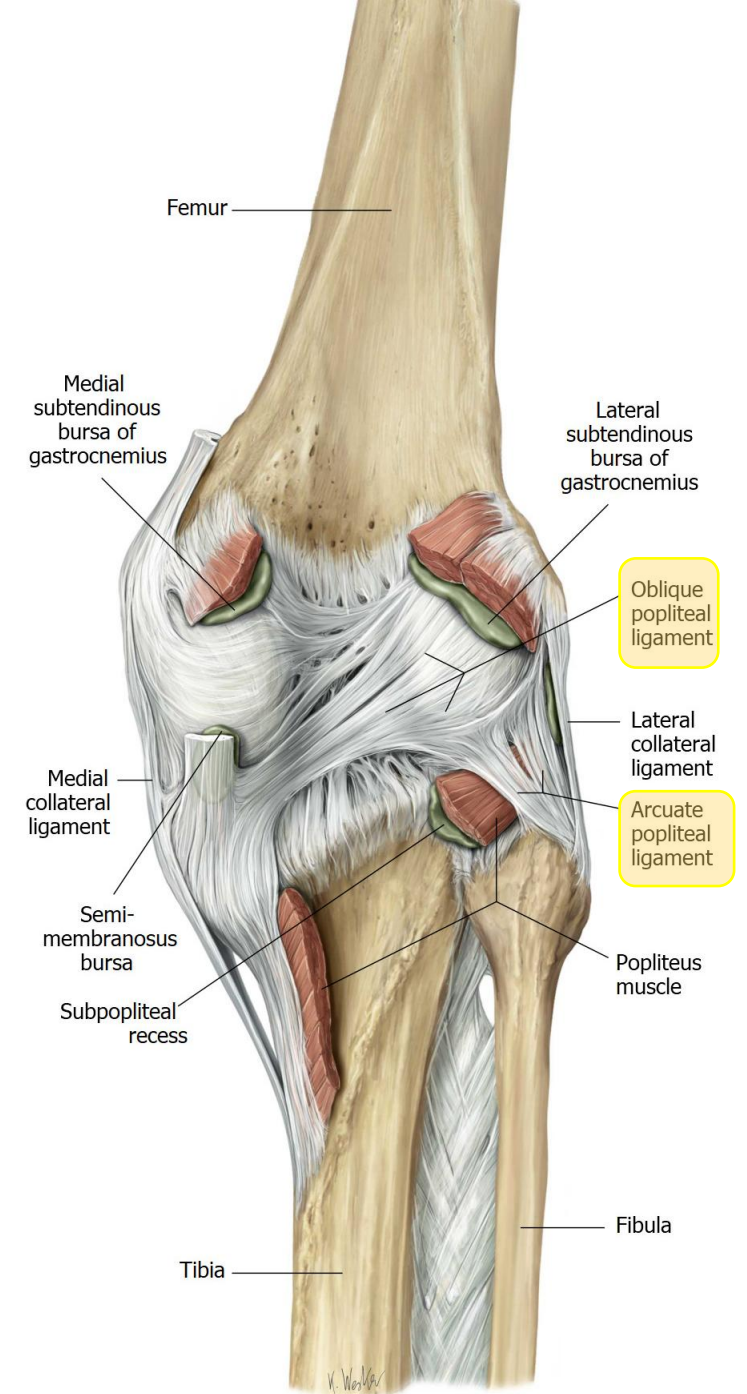
Oblique and Arcuate Ligaments

Oblique popliteal ligament

- runs from medial tibial condyle to lateral femoral condyle
- expansion of semimembranosus muscle
- reinforces joint capsule; prevents hyperextension and external rotation of the knee

Arcuate popliteal ligament

- runs from fibular head to superomedial posterior knee joint capsule
- provides stability for the posterior knee and joint capsule



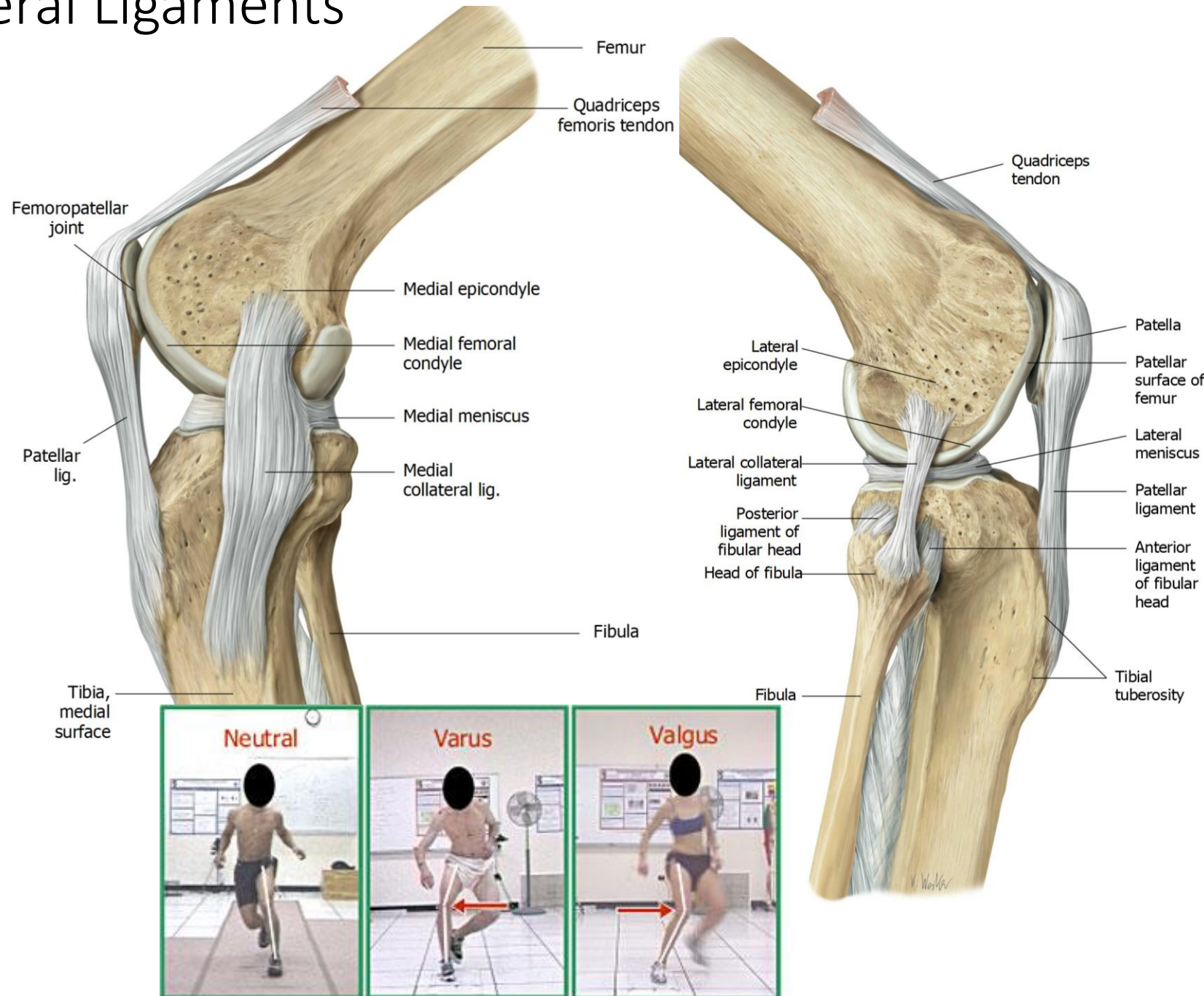
Fibular and Tibial Collateral Ligaments

Fibular (lateral) collateral ligament

- cord-like from lateral femoral epicondyle to fibular head
- no meniscal attachment
- limits *varus force* and axial rotation with knee flexed

Tibial (medial) collateral ligament

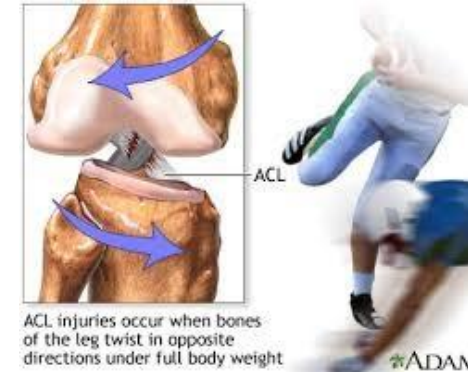
- broad, flat ligament from medial femoral epicondyle to proximal tibia
- has extracapsular portion
 - vertical fibers that are anterior and superficial to joint capsule
- Has capsular portion
 - Oblique fibers that attach to joint capsule and medial meniscus
- Resists *valgus force* and limits axial rotation when knee is flex



Intracapsular Ligaments

Anterior cruciate ligament

- Attaches from anterior intercondylar area of tibial plateau to medial side of the lateral femoral condyle
- Resists excessive anterior translation of the tibia when femur is fixed or posterior translation of femur on fixed tibia
- Resists hyperextension of knee



Posterior cruciate ligament

- Attaches from posterior intercondylar region of tibia to lateromedial femoral condyle
- Limits excessive posterior translation of tibia on fixed femur or anterior translation of femur on fixed tibia

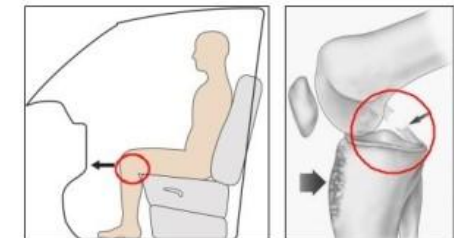
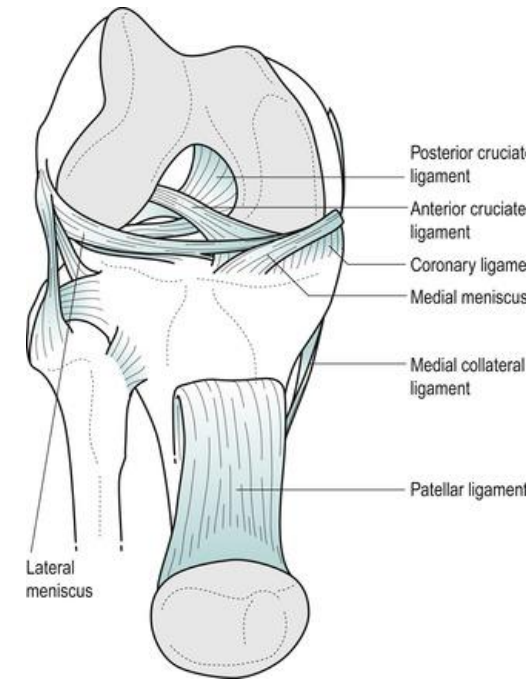
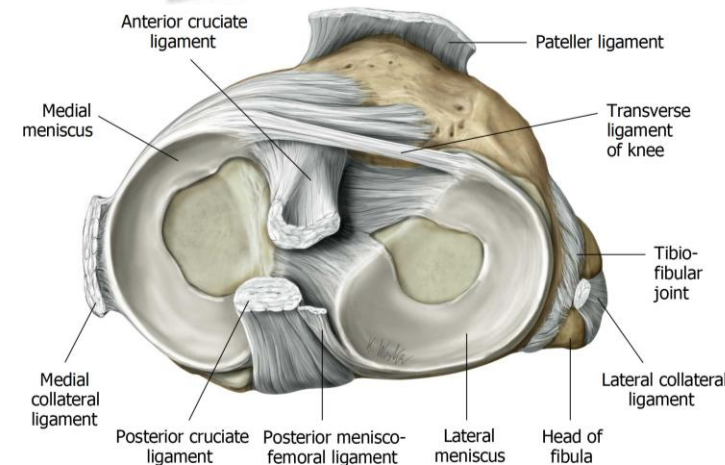


Medial and Lateral Menisci

- Wedge-shaped fibrocartilaginous discs
- Improve joint congruence
- Act as shock absorbers and limit hyperextension

Coronary ligaments

- Attach to menisci to tibia
- Allow menisci to pivot



Dashboard injury

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